Equation Sample 1

Dataset-expansion sample

1 Derivation

The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$
\begin{aligned}
\alpha_3 &\overset{\text{(i)}}{\leq} \sum_{s\in\mathcal{S},a\in\mathcal{A}} d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)\sqrt{\frac{C^\star_{\mathsf{clipped}}\log\frac{N}{\delta}}{N\min\Big\{d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big),\frac{1}{S(A+B)}\Big\}}\mathsf{Var}_{P_{s,a,\nu_0(s)}}\big(V^-_{\mathsf{pe}}\big)} \\
&\leq \sqrt{\frac{C^\star_{\mathsf{clipped}}\log\frac{N}{\delta}}{N}}\sum_{s\in\mathcal{S},a\in\mathcal{A}}\sqrt{d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)\mathsf{Var}_{P_{s,a,\nu_0(s)}}\big(V^-_{\mathsf{pe}}\big)} \\
&\quad+\sqrt{\frac{C^\star_{\mathsf{clipped}}S\left(A+B\right)\log\frac{N}{\delta}}{N}}\sum_{s\in\mathcal{S},a\in\mathcal{A}}\sqrt{d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)}\cdot\sqrt{d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)\mathsf{Var}_{P_{s,a,\nu_0(s)}}\big(V^-_{\mathsf{pe}}\big)} \\
&\overset{\text{(ii)}}{\leq} \sqrt{\frac{C^\star_{\mathsf{clipped}}}{N}\log\frac{N}{\delta}}\cdot\sqrt{SA}\cdot\sqrt{\sum_{s\in\mathcal{S},a\in\mathcal{A}}d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)\mathsf{Var}_{P_{s,a,\nu_0(s)}}\big(V^-_{\mathsf{pe}}\big)} \\
&\quad+\sqrt{\frac{C^\star_{\mathsf{clipped}}S\left(A+B\right)\log\frac{N}{\delta}}{N}}\left[\sum_{s\in\mathcal{S},a\in\mathcal{A}}d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)\right]\sqrt{\sum_{s\in\mathcal{S},a\in\mathcal{A}}d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)\mathsf{Var}_{P_{s,a,\nu_0(s)}}\big(V^-_{\mathsf{pe}}\big)} \\
&\leq 2\sqrt{\frac{C^\star_{\mathsf{clipped}}S\left(A+B\right)\log\frac{N}{\delta}}{N}}\sqrt{\sum_{s\in\mathcal{S},a\in\mathcal{A}}d^{\mu^\star,\nu_0}\big(s,a,\nu_0(s);\rho\big)\mathsf{Var}_{P_{s,a,\nu_0(s)}}\big(V^-_{\mathsf{pe}}\big)} \\
&\overset{\text{(iii)}}{=} 2\sqrt{\frac{C^\star_{\mathsf{clipped}}S\left(A+B\right)}{N}\log\frac{N}{\delta}}\sqrt{\sum_{s\in\mathcal{S}}d^{\mu^\star,\nu_0}\left(s;\rho\right)\mathop{\mathbb{E}}_{a\sim\mu^\star(s),b\sim\nu_0(s)}\Big[\mathsf{Var}_{P_{s,a,b}}\big(V^-_{\mathsf{pe}}\big)\Big]} \\
&\overset{\text{(iv)}}{\leq} 2\sqrt{\frac{C^\star_{\mathsf{clipped}}S\left(A+B\right)}{N}\log\frac{N}{\delta}}\sqrt{\sum_{s\in\mathcal{S}}d^{\mu^\star,\nu_0}\left(s;\rho\right)\mathsf{Var}_{P_s^{\mu^\star,\nu_0}}\big(V^-_{\mathsf{pe}}\big)}.
\end{aligned}
$$